



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Civil Engineering
In-Semester Examination – I (ISE-I)
Academic Year: 2025-26 (Semester I)

Class: Third Year (TY) B.Tech.

Course: Transportation Engineering Course Code: PCC-CE304

Date: 21.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. Mobile phones and programmable calculators are strictly prohibited.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. Figures to right indicate full marks.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain in detail the road patterns with neat sketch.	05	1	1
Q.2	Explain the first and second 20-year road development plan in detail.	05	1	2
Q.3	Calculate the safe stopping sight distance in m for design speed of 50 km/hr for two-way traffic on a two-lane road and two-way traffic on a single lane road. The brake efficiency is 80 % and longitudinal coefficient of friction is 0.35.	05	2	3
Q.4	A ascending gradient of 1 in 50 meets a descending gradient of 1 in 80. A summit curve is to be designed for a speed of 80 km/hr to have an intermediate sight distance of 280 m.	05	2	3



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Department of Civil Engineering
In-Semester Examination - I
Academic Year: 2025-26 (semester I)

Class: Third Year (TY) B.Tech.

Course: Geotechnical Engineering

Course Code: PCC-CE303

Date: 18.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. *Mobile phones and programmable calculators are strictly prohibited.*
- ii. *Exchange/Sharing of stationery, calculator etc. not allowed.*
- iii. *Figures to right indicate full marks.*

Q.N.	Questions	Marks	CO	BT
Q.1	State the formula to find out the particle size range and shape of the grain size distribution curve of soil.	02	CO1	BT1
Q.2	Define water content, void ratio, porosity and degree of saturation with formula.	04	CO1	BT2
Q.3	Explain in details dispersed and flocculated type of clay structure.	06	CO1	BT2
Q.4	A saturated sample of soil has a water content of 35.5% and specific gravity of 2.70. Calculate dry density, saturated density and submerged density in kN/m^3 .	08	CO1	BT3



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded		
ISE-I (Semester-I) 2025-26		
Third Year B. Tech.	Civil Engineering	
Subject: Building Materials and Concrete Technology	Subject Code: PCC-CE301	
Date: 22-Aug-2025	Time: 45 Minutes (15:00 to 15:45)	Max. Marks: 20

Q. No.	Questions (Solve any TWO)	Marks	CO	BT
Q. 1)	Explain the wet process of portland cement manufacturing with schematic layout.	10	2	1
Q. 2)	Enlist and explain any FIVE types of cements.	10	1	1
Q. 3)	Explain the laboratory method for determining the initial and final setting time of the cement with neat sketch of apparatus.	10	2	2

(P.T.O.)



Class: TY

Course: Design of Steel Structures Course Code: PCC-CE301

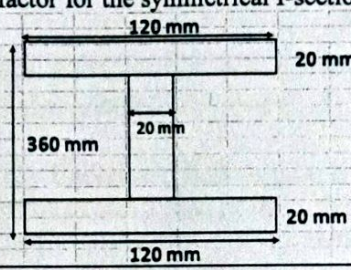
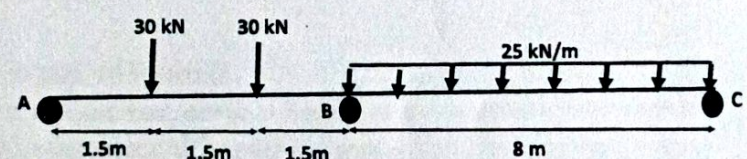
Date: 20.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Advantages of steel over concrete in construction practice.	3	CO1	BT1
Q.2	Determine the shape factor for the symmetrical I-section shown in fig. 	6	CO3	BT4
Q.3	A continuous beam is subjected to a collapse load system with roller support, as shown in Fig. If the beam is of uniform cross-section throughout its length, determine the plastic moment capacity required for the beam. 	6	CO3	BT4
Q.4	Discuss failure modes in structural steel connections (per IS 800-2007)	5	CO4	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate an understanding of the material properties of steel and their behaviour under various load conditions to make design decisions.
CO 2	Design a steel structural component (beams, columns, connections) according to design codes and standards.
CO 3	Analyze steel structures to determine internal forces, moments, and shear, and apply the appropriate methods for different loading scenarios.
CO 4	Design safe and effective bolted and welded connections for steel structures, ensuring structural integrity and efficiency.
CO 5	Effectively interpret and apply steel design codes and practical implementation of designs



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
In semester Examination 1
Academic Year: 2025-26

Class: TY	Subject: Engineering Geology	Subject Code: AUD-CE308
Date: 22.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. Mobile phones and programmable calculators are strictly prohibited.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.
- v. All questions are compulsory

Q.N.			Marks	CO	BT
Q. 1	a	What is weathering? explain types of weathering.	10	CO1	BT2
Q. 2	a	Write short note on metamorphic rock and igneous rock	10	CO2	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand weathering process and mass movement
CO 2	Distinguish geological formations
CO 3	Identify geological structures and processes for rock mass quality
CO 4	Identify subsurface information and groundwater potential sites through geophysical investigations
CO 5	Apply geological principles for mitigation of natural hazards and select sites for dams and tunnels



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Civil Engineering ISE-I Examination Academic Year: 2025-26 (Semester I)

Class: T.Y. B. Tech.
Date: 28/08/2025

Subject: Engineering Hydrology
Time: 03:00 p.m. to 03:45 p.m.

Subject Code: PCC-CE303
Max. Marks: 20

Note: i. Assume suitable data whenever necessary and state it clearly. ii. Mobile phones and programmable calculators are strictly prohibited. iii. Exchange/Sharing of stationery, calculator etc. not allowed. iv. Figures to right indicate full marks. v. The symbols carry usual meaning.

Q. No.	Question	Marks	CO	BT
Q.1(a)	What are the different forms of precipitation? The normal annual rainfall at stations A, B, C and D in a basin are 80.97, 67.59, 76.28 and 92.01 cm respectively. In a particular year, the station D was inoperative and the stations A, B and C recorded annual precipitation of 91.11, 72.23 and 79.89 cm respectively. Estimate the rainfall at station D in that year.	5	CO1	BT1
Q.1(b)	Explain the Double-mass curve technique for checking consistency of rainfall data.	5	CO1	BT2
Q.2(a)	Discuss the factors that affect evaporation from a water body.	5	CO2	BT1
Q.2(b)	The infiltration capacity in a basin is represented by Horton's equation as $f_p = 3.0 + e^{-2t}$. Estimate the depth of infiltration in first 30 minutes. Take the duration of storm as 60 minutes.	5	CO2	BT2

Bloom's Taxonomy (BT) : BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create
Course Outcomes: On successful completion of this course, students should be able to:
CO 1: Analyse precipitation data. CO 2: Understand evaporation and infiltration processes.